

Question ID: 113b938e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question
 $y = 18 - 5x$

The equation above represents the speed y , in feet per second, of Sheila’s bicycle x seconds after she applied the brakes at the end of a ride. If the equation is graphed in the xy -plane, which of the following is the best interpretation of the x -coordinate of the line’s x -intercept in the context of the problem?

- Answer
- A. The speed of Sheila’s bicycle, in feet per second, before Sheila applied the brakes
 - B. The number of feet per second the speed of Sheila’s bicycle decreased each second after Sheila applied the brakes
 - C. The number of seconds it took from the time Sheila began applying the brakes until the bicycle came to a complete stop
 - D. The number of feet Sheila’s bicycle traveled from the time she began applying the brakes until the bicycle came to a complete stop

Correct Answer: C

Rationale

Choice C is correct. It’s given that for each point (x, y) on the graph of the given equation, the x -coordinate represents the number of seconds after Sheila applied the brakes, and the y -coordinate represents the speed of Sheila’s bicycle at that moment in time. For the graph of the equation, the y -coordinate of the x -intercept is 0. Therefore, the x -coordinate of the x -intercept of the graph of the given equation represents the number of seconds it took from the time Sheila began applying the brakes until the bicycle came to a complete stop.

Choice A is incorrect. The speed of Sheila’s bicycle before she applied the brakes is represented by the y -coordinate of the y -intercept of the graph of the given equation, not the x -coordinate of the x -intercept. Choice B is incorrect. The number of feet per second the speed of Sheila’s bicycle decreased each second after Sheila applied the brakes is represented by the slope of the graph of the given equation, not the x -coordinate of the x -intercept. Choice D is incorrect and may result from misinterpreting x as the distance, in feet, traveled after applying the brakes, rather than the time, in seconds, after applying the brakes.